CleLight protocol – Brain clearing

Bench protocol ~ 2-3 weeks

All incubations with shaking. The volume of solution must be **at least** 3 times the volume of the tissue and the sample must not be squeezed in the tube.

⚠ *Wear PPE. Manipulations with DCM or DBE to be done under a fume hood.*
Use Polypropylene (PP) tubes or glass vials for DCM and DBE incubations.

Methanol test of antibodies

Make sure antibodies are compatible with methanol treatment. To do so, dehydrate progressively tissue sections (FFPE or frozen sections) and incubate in methanol 10min. Rehydrate progressively and process the immunolabeling as usual. If the labeling has a good signal to noise ratio, the antibody is compatible with methanol treatment.

Antigen retrieval (optional)

For tissue stored in fixative for more than 5 years and specific antibodies.
Heat to 50kPa for 5min in **citrate buffer** in a pressure cooker.

Sample Pretreatment

Wash well tissue with PTw overnight if it was stored in PFA.

1. Dehydrate with methanol/H_2O (v/v): 50%, 80%, 2x100%, 1h each RT
2. Incubate in 100% methanol 1h +4°C
3. Incubate in 66% DCM / 33% methanol (v/v) overnight RT. *Sample should fall at bottom of vial. If not, wait longer.*
4. Incubate in 2x 100% methanol 30min RT
5. Incubate in **Fresh 5% H_2O_2** (v/v) in 100% methanol, overnight at 4°C
6. Rehydrate with methanol/H_2O (v/v): 80%, 50%, then PTx 1h each. *Tissue sinks in tube when hydrated.*

7. Incubate in **Fresh permeabilization solution** at 37°C for 3 days. Refresh the solution every day.
8. Wash 2x 30min with PTw.

9. Incubate in **Quadrol (25%) / ammonium (5%)** overnight at 37°C.
10. Wash 5x 30min with PTw.

1st Clearing

1. Dehydrate in methanol/H_2O (v/v): 50%, 80%, 2x 100%, 1h each at RT. *Sample can be left overnight in 100% methanol for better dehydration*.
2. Incubate in 66% DCM / 33% methanol 3h at RT. *Sample should sink.*
3. Incubate 100% DCM 2x 15min at RT

4. Transfer to Dibenzyl Ether (DBE) at RT. *Fill the tube completely to prevent sample oxidation.*

Light clearing and rehydration

1. After sample is cleared, leave in DBE **under LED light** for at least 2 days
2. Incubate in 100% DCM 2x 15min at RT
3. Incubate 66% DCM / 33% methanol 3h at RT
4. Rehydrate with methanol/H_2O (v/v): 100%, 80% and 2x 50%, 1h each at RT
5. Put in PTw and let sample rehydrate overnight at RT

ECM loosening

1. Incubate overnight in 0.5M **acetic acid** diluted in H_2O at 37°C
2. Incubate overnight in **Guanidine hydrochloride solution** at 37°C
3. Wash 5x 30min in PTw

Immunolabeling

1. Incubate with **Blocking Solution** overnight at 37°C
2. Incubation with **primary antibody** in **antibody dilution buffer** at 37°C for 7 days. *Centrifuge before incubation to remove potential antibody aggregates.*
3. Wash 5x in PTw over 1 day
   Incubation with **secondary antibody** in PTw at 37°C for 7 days *Protect your sample from the light from this point until the end of the protocol.*
4. Small molecule dyes (optional) e.g. To-Pro-3, Biotracker green, Labeled Lectin: diluted in **PBS with high Na** solution and incubate for 5-7 days at 37°C

   Option for faster labeling:
5. Combine primary + secondary Fab antibodies. Preincubate for 30min in **antibody dilution buffer** before incubating the sample for 7 days. *Centrifuge before incubation to remove potential antibody aggregates.*

6. Wash 5x in PTw over 1 day

Final clearing

1. Dehydrate in methanol/H_2O series: 50%, 80%, 2x100%, 1h each at RT. *Sample can be left overnight in 100% methanol for better dehydration*. G*ood dehydration is essential for good clearing.*
2. Incubate in 66% DCM / 33% Methanol 3h at RT. *Sample should sink.*
3. Incubate in 100% DCM 2x at RT for 15 min each
4. Incubate in Dibenzyl Ether (DBE) at RT. *Fill the tube to the tope prevent oxidation.*

Imaging

After imaging, samples can be photobleached using the **LED illuminator** for 5 to 7 days at 10°C. Tissue is then rehydrated and processed though a second round of labeling (for immunostaining, make sure there is no cross-talk).

Materials

Acetic acid	302002 Carlo Erba
Ammonium hydroxide solution (~25% NH_3 basis, NH_4OH)	30501-1L Sigma-Aldrich
Citric acid monohydrate	1.00244.0500 Merck
Dibenzyl ether (DBE)	33630 Sigma-Aldrich
Dichloromethane (DCM)	D/1852/17 Fisher Chemical
Dimethyl sulfoxide (DMSO)	41640 Sigma-Aldrich
Gelatine	24350.262 VWR Chemicals
Glycine	A1067 PanReac Applichem
Guanidine hydrochlorid	6069.3 Carl Roth
Heparin	2022511 Braun
Hydrogen peroxide 35% w/w (H_2O_2)	L14000.AP Thermo Scientific
Methanol	M/4000/17 Fisher Chemical
Phosphate buffered saline (PBS)	P4417-100TAB Sigma-Aldrich
Potassium chloride (KCl)	P4504-500G Sigma-Aldrich
Potassium Phosphate Monobasic	P-5379 Sigma-Aldrich
Quadrol (N,N,N', N'-Tetrakis(2-hydroxzpropyl)ethylenediamine)	122262-1L Sigma-Aldrich
Saponine	S4521 Sigma-Aldrich
Sodium acetate	32319 Sigma-Aldrich
Sodium azide	71289 Sigma-Aldrich
Sodium chloride (NaCl)	S9888-25G Sigma-Aldrich
Sodium hydroxide (NaOH)	480501 Carlo Erba
Sodium Phosphate Dibasic	71640-1KG Sigma-Aldrich
Triton X-100	T9284 Sigma-Aldrich
Tween 20	A4974 PanReac Applichem

Equipment

White LED projector	15000 lm, 4000 K neutral white, Hornbach, Switzerland

Solutions

PBS Tween 0.2 % (PTw)

For 1L:

- 2 mL Tween 20 (0.2 %)
- 0.2 g Sodium azide (0.02 %)
- PBS 1X up to 1 L

PBS Triton X-100 0.3 % (PTx)

For 1L:

- 3 mL Triton X-100 (0.3 %)
- 0.2 g Sodium azide (0.02 %)
- PBS 1X up to 1 L

Citrate Buffer pH 6.0:

For 1L:

- 1.92 g Citric acid
- 0.5 mL Tween 20 (0.05 %)
- Distilled water up to 1L

Dissolve citric acid in water. Adjust pH to 6.0 with 1N NaOH. Add 0.5 ml of Tween 20 and mix well. Store at 4°C.

Permeabilization solution

For 1L:

- 30 mL Triton X-100 (0.3%)
- 23 g Glycine
- 10 g Saponine (1 %)
- 0.2 g Sodium azide (0.02 %)
- 200 mL DMSO (20%)
- PBS 1X up to 1 L

Prepare fresh – store at 4°C for max 1-2 month

25% Quadrol - 5% Ammonium Solution

For 1L:

- 250 mL of Quadrol
- 200 mL NH_4OH 25 % solution
- Water up to 1 L

Quadrol has viscous consistency

0.5 M Acetic acid

For 1L:

- 28,62 mL Acetic acid
- Water up to 1 L

Add acetic acid to water, then top up to 1 L

Wear protective googles

Guanidine HCl solution

For 50 mL:

- 19,1 g Guanidine hydrochloride (4M)
- 0,21 g Sodium acetate (0.05 M)
- 1,5 mL Triton X-100 (3 %)
- PBS 1X up to 50 mL

Prepare fresh

Blocking Solution

For 1L:

- 0.2 g Sodium azide (0.02 %)
- 2 g Gelatine (0,2 %)
- 100 mL DMSO
- 840 mL PBS 1X

Heat the PBS-Azide. Add gelatin, wait until it melts. Then add DMSO.

PBS with high Na (500 mM NaCl)

For 1L:

- 29,22 g NaCl
- 0,2 g KCl
- 1,44 g Sodium Phosphate Dibasic
- 0,245 g Potassium Phosphate Monobasic
- 0.2 g Sodium azide (0.02 %)
- PBS 1X up to 1L

Antibody dilution buffer

For 1 L:

- 2 mL Tween 20 (0.2 %)
- 0.2 g Sodium azide (0,02 %)
- 10 mg/ml Heparin
- 50 mL DMSO (5%)
- PBS 1X up to 1 L